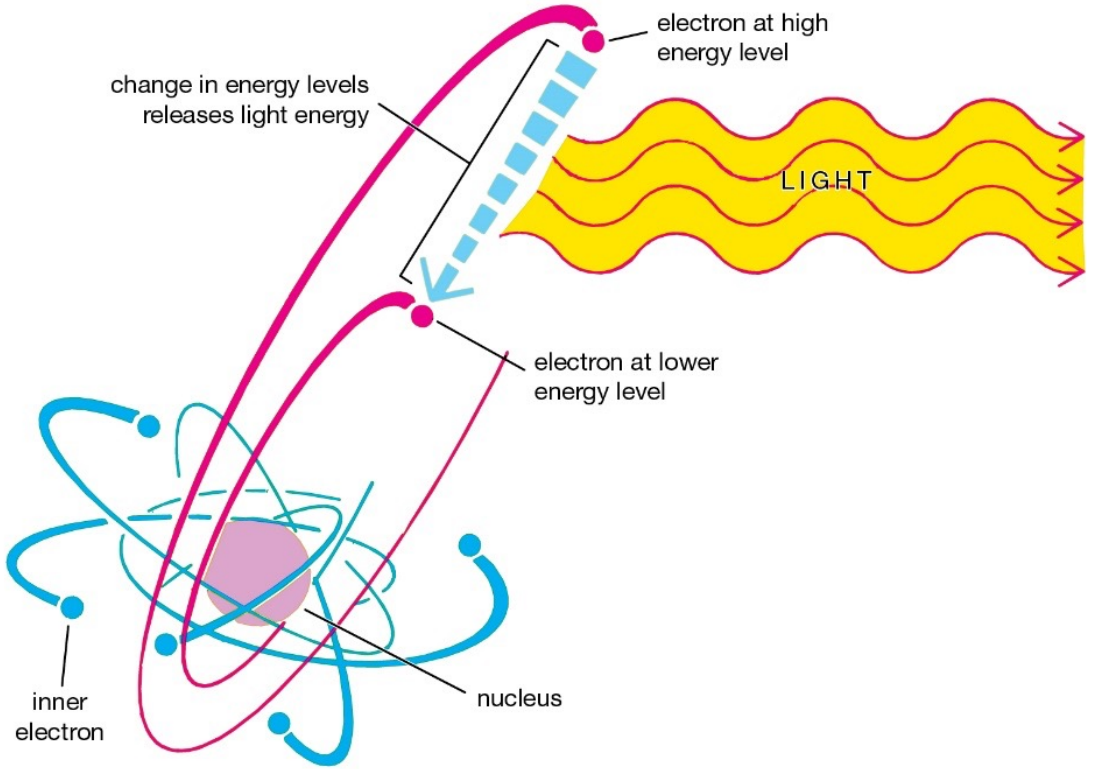
Physics 365:
Acoustics, Optics and Modern Physics

Week 9: Light Matter Interaction

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University of Calgary

Fall 2026



UNIVERSITY OF CALGARY

EMISSION AND ABSORBTION OF LIGHT

Question 1

The total Power of a source of sound waves is P. If I add a second speaker with the same power output. How much would the Power of the oscillating particles increase?

A. The power will double.

B. The power will remain the same.

C. The power will decrease by half.

D. It depends on the frequency of oscillations.

Historical context

- **Rayleigh–Jeans law:** in summer of 1900

$$B_\lambda(T) = \frac{2cK_BT}{\lambda^4}$$

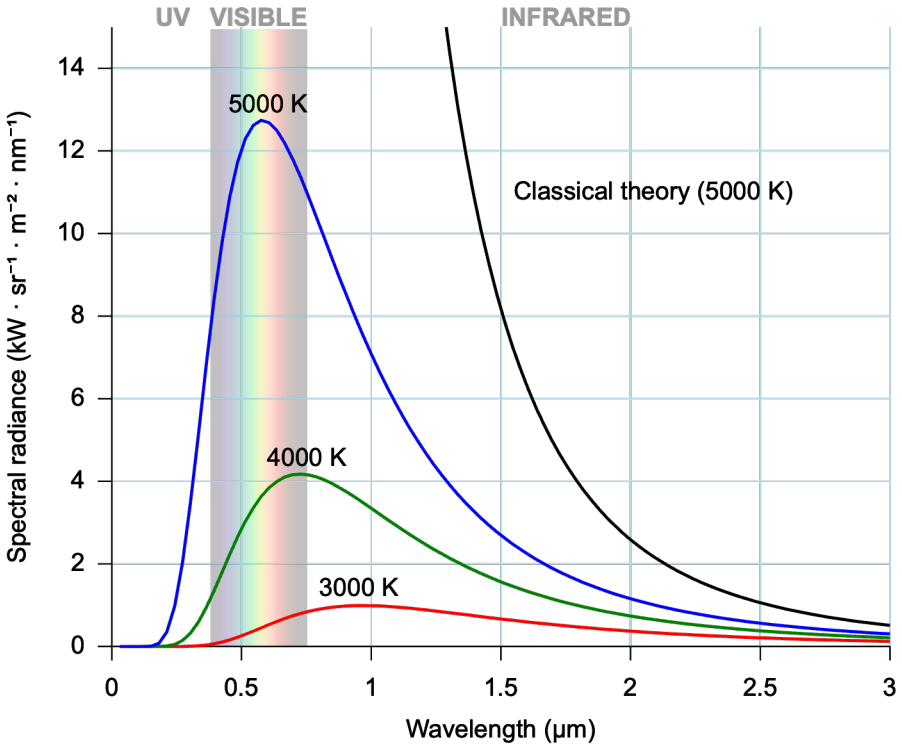
- **Planck Postulate:** The **Energy of an Oscillator** can only Increase and Decrease by **Discrete portions of Energy**

$$E = hf$$

where $h = 6.63\times10^{-34}\text{J}.s$ is the **Planck's constant**

- **Planck's Law:** winter of 1900

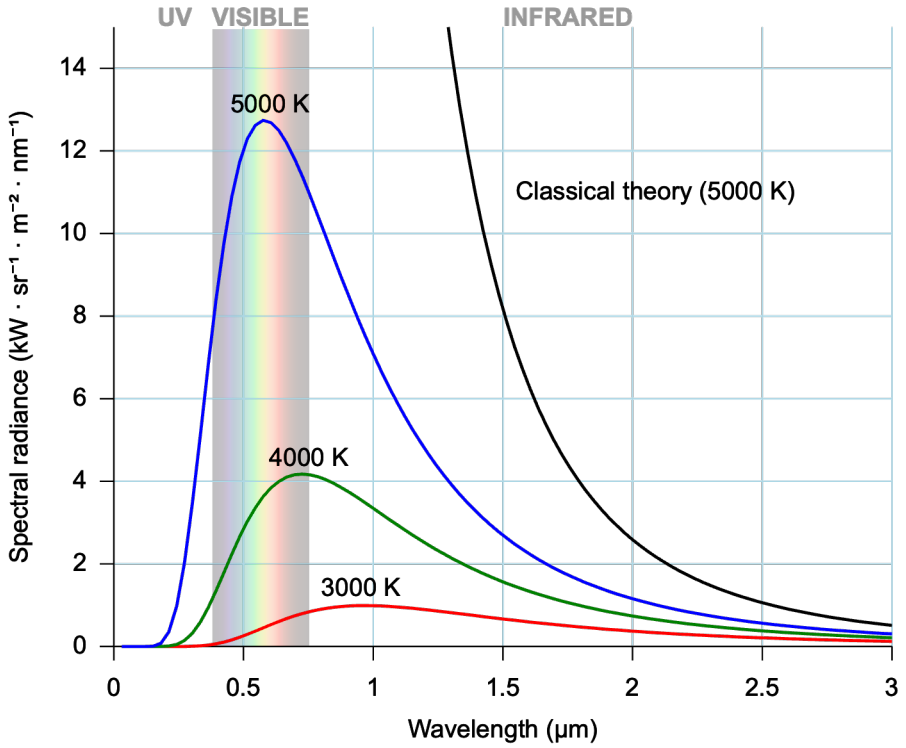
$$B_\lambda(T) = \frac{2c^2h}{\lambda^5}\frac{1}{e^{hc/(\lambda K_BT)}-1}$$



By Darth Kule - Own work, Public Domain, https://commons.wikimedia.org/w/index.php?curid=10555337

Historical context

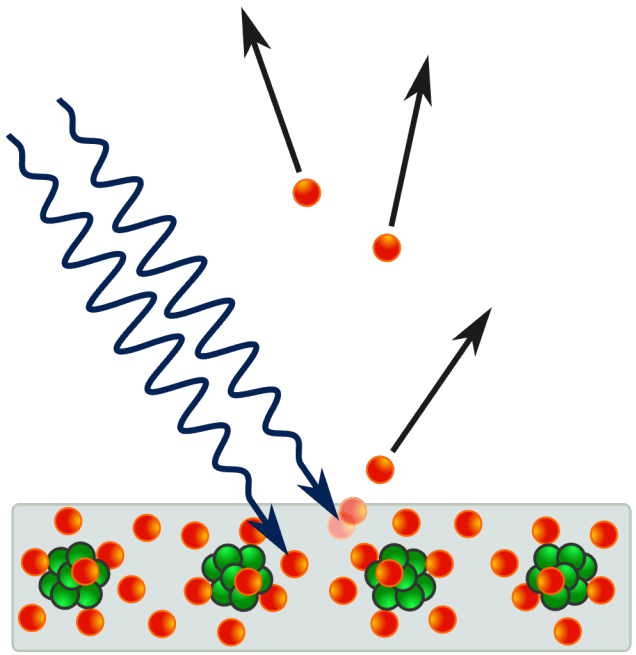
- **Planck** regarded the discretisation of absorption and emission of light as a mathematical trick. Thus, the electromagnetic wave fundamentally to be a **classical wave of continuous media**.
- **Albert Einstein** thought it a fundamental law of nature that a beam of light is a constant stream of localized packets of energy named **Photons** by Gilbert N. Lewis in 1926.



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The Classical Description

- **According to Classical Description:** the energy carried by the Light will be spent on the amount of energy needed to separate the electron from the metal lattice called a **Work Function** Φ of a material and the leftover will be given to the electron in the form of **Kinetic Energy** K_{max}



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Question 2

What will happen in the Classical picture if we increase the intensity of the incident light?

more I → more E

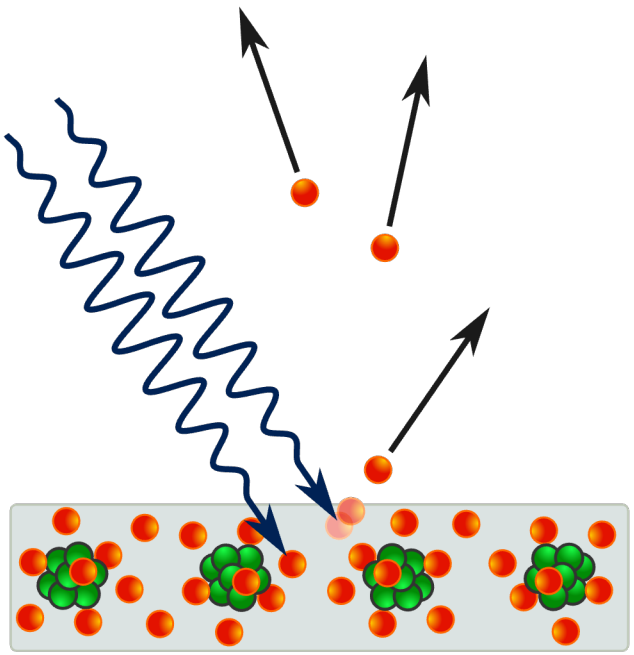
$E_{light} = \phi + k_{electron}$

A. Nothing.
B. The kinetic energy of the freed electrons will increase.
C. The kinetic energy of the freed electrons will decrease.
D. The remaining energy will be reflected back in the firs medium.

The Classical Description

- **According to Classical Description:** the energy carried by the Light will be spent on the amount of energy needed to separate the electron from the metal lattice called a **Work Function** Φ of a material and the leftover will be given to the electron in the form of **Kinetic Energy** K_{max}
- According to classical physics Increasing the intensity of the incident light will increase the kinetic energy of the freed electrons.

WILL OBSERVATIONS CONFIRM?



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First experiment of photoelectric effect

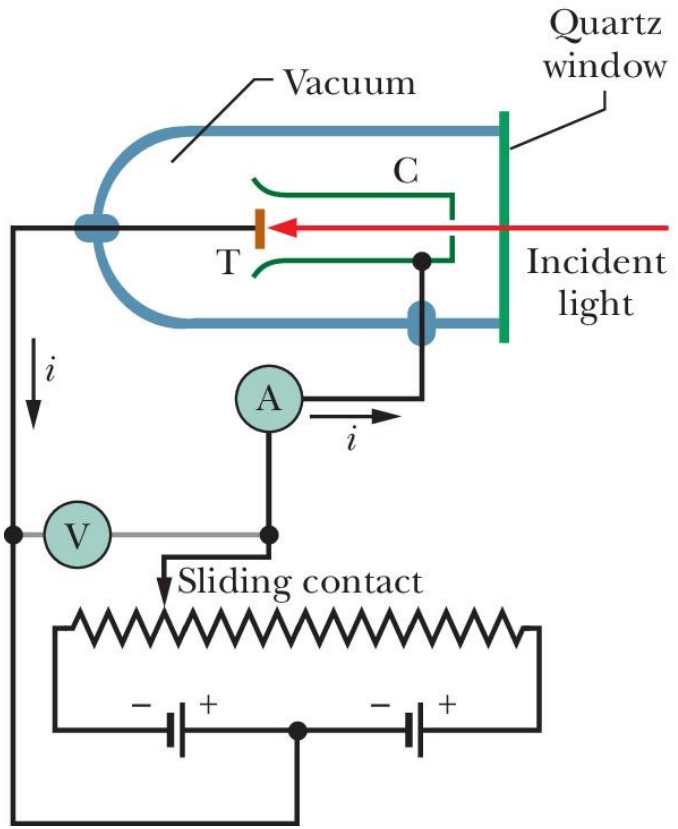
- We vary the **Voltage Difference** until the **Current** becomes **Zero**.
- The kinetic energy of the freed electrons then will be

$$K_{max} = eV_{stop}$$

where V_{stop} is called the **Stopping Voltage**.

- Then **Intensity of the** Light is Increased and **No Change in the Stopping Volage** was observed

CONTRADICTION WITH CLASSICAL PHYSICS!



Second experiment of photoelectric effect

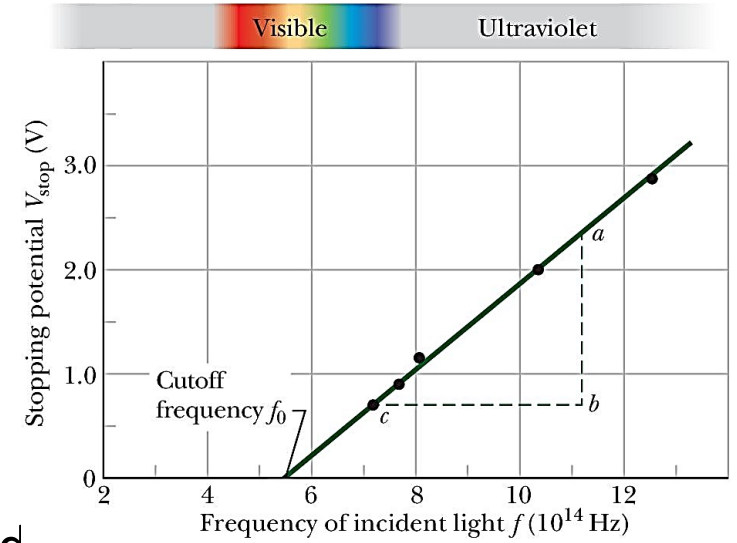
- We Fix the **Voltage Difference** and vary the **Frequency** f of the Light
- The observations show that below a certain **Frequency** f_0 the Current stops.
- To just escape the material and have no additional **Kinetic Energy** the **Energy of the Photons** should be equal to the **Work Function**

$$hf_0 = \Phi$$

- In 1905 **Albert Einstein** proposed a theoretical description of the experimental results observed by P. Lenard, J. J. Thomson, and F. Palmer Jr.

Electrons can escape only if the light frequency exceeds a certain value.

The escaping electron's kinetic energy is greater for a greater light frequency.



Photoelectric Effect

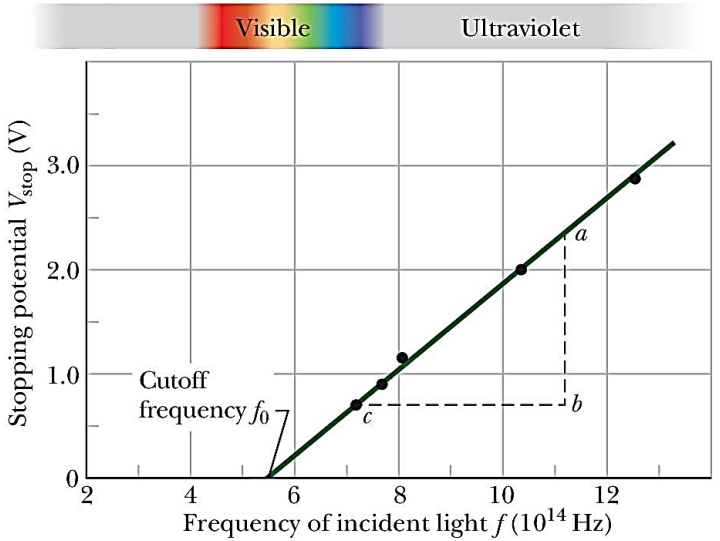
- The mathematical description of the Photoelectric effect is as follows

$$hf = K_{max} + \Phi \quad (= eV_{stopping})$$

- **Energy of a single photon**
- **Maximum Kinetic Energy acquired by the freed electrons**
- **Work function: The energy required to free an electron from the surface of the Metal**

Electrons can escape only if the light frequency exceeds a certain value.

The escaping electron's kinetic energy is greater for a greater light frequency.



Question 3

The graph on the right is taken from R. A. Millikan's Noble prize-winning paper published in 1916. What is the work function of the metal he used?

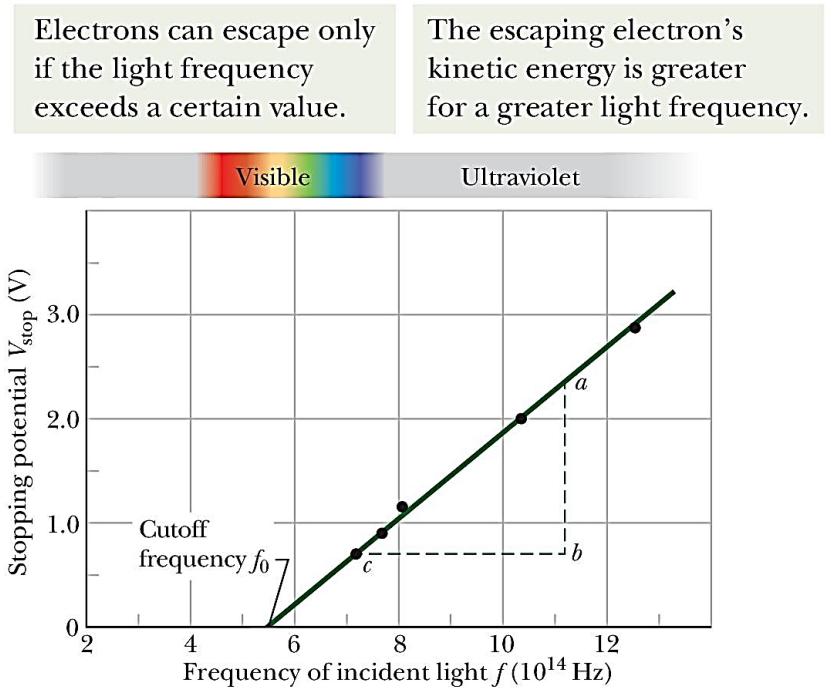
The electron charge is $1.60217663\times10^{-19}C$.

A. $3.6465\times10^{-19}J$

B. $1.6021\times10^{-19}J$

C. $2.276J$

D. $0J$



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PHOTON MOMENTUM

Photon Momentum

- **Particle Momentum:** A quantity describing the amount of motion in a particle.

$$\vec{p} = m\,\vec{v}$$

$$E \propto m\vec{v} \cdot \vec{v}$$

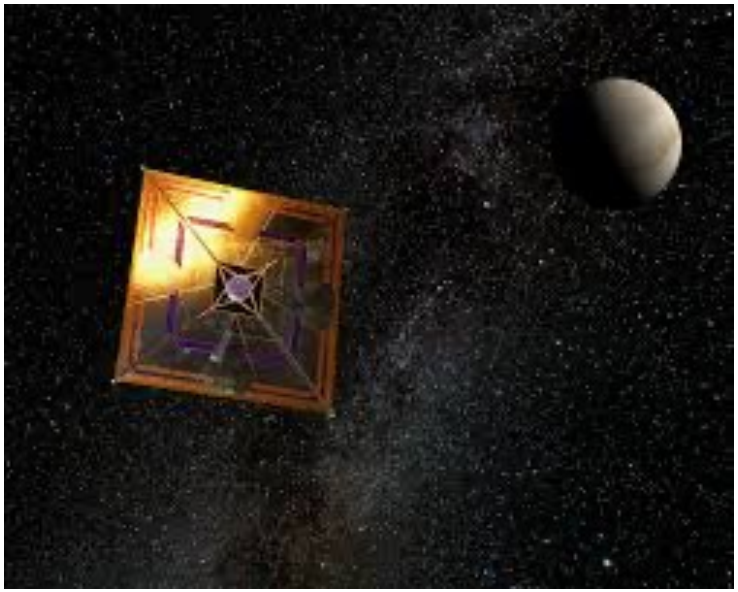
$$\vec{p} \propto \frac{E}{\vec{v}}$$

- We know the Energy of the photon is

$$E = hf$$

- **Photon Momentum:**

$$\vec{p} = \frac{hf}{c}\,\hat{s} = \frac{h}{\lambda}\,\hat{s}$$



https://en.wikipedia.org/wiki/Solar_sail

Special Relativity Context

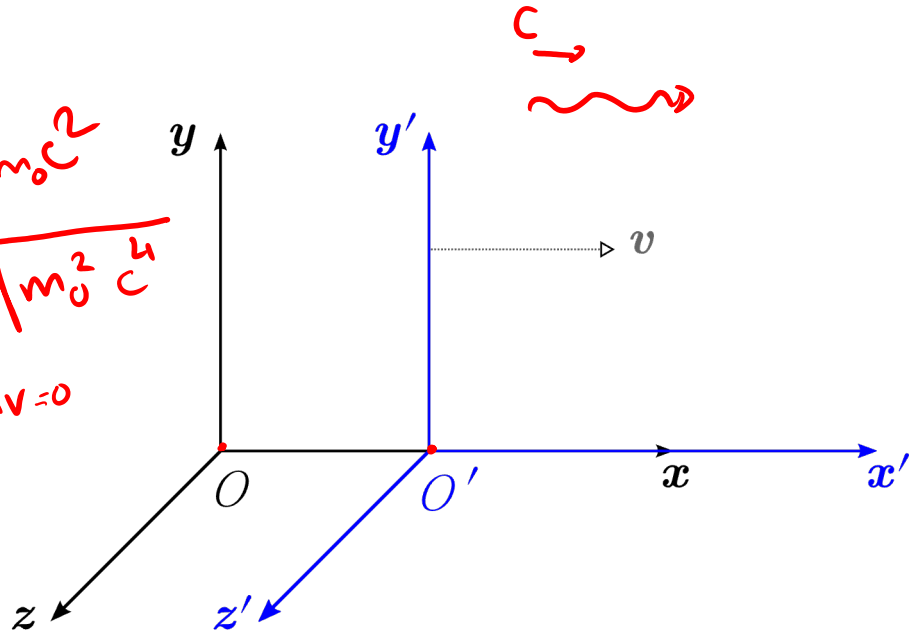
- **Frames of reference:**

$$t' = \gamma\left(t - \frac{vx}{c^2}\right)$$
$$x' = \gamma(x - vt)$$
$$y = y'; z = z'$$
$$\gamma = 1/\sqrt{1 - v^2/c^2}$$

- **Special Relativity:** Einstein's dispersion relation

$$E = \sqrt{p^2c^2 + m_0^2c^4}$$

- m_0 is the **Rest Mass** and since photons are massless particles: $m_0 = 0$



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Classical Conservation of Momentum And Energy

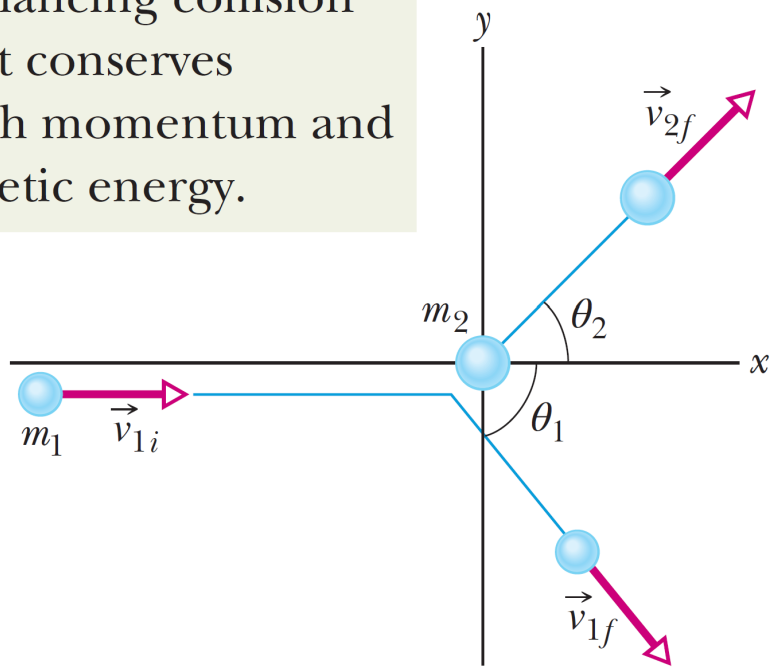
- **Momentum Conservation:** The total **Momentum** (both **Magnitude** and **Direction**) of the system through elastic collisions does not change.

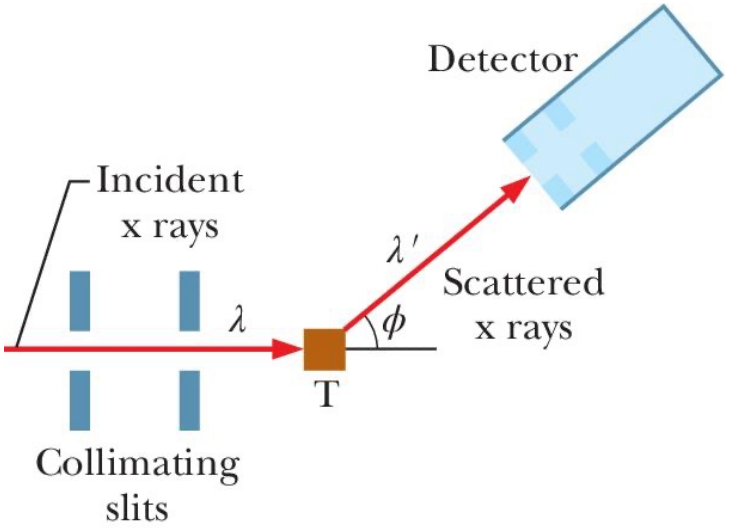
$$m_1\vec{v}_{1i} = m_2\vec{v}_{2f} + m_1\vec{v}_{1f}$$

- **Energy Conservation**: Energy does not change in a system without friction.

$$\frac{1}{2}m_1v_{1i}^2 = \frac{1}{2}m_2v_{2f}^2 + \frac{1}{2}m_1v_{1f}^2$$

A glancing collision that conserves both momentum and kinetic energy.

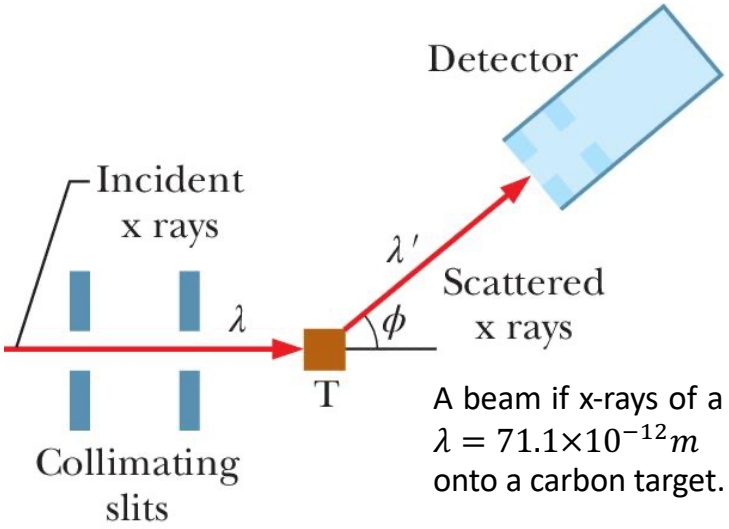




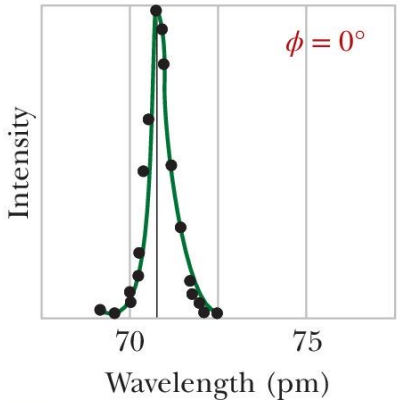
COMPTON EXPERIMENT

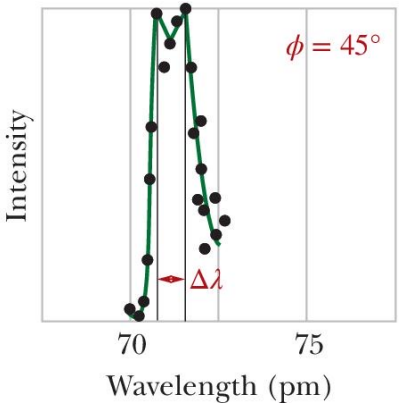
Compton's Experiment

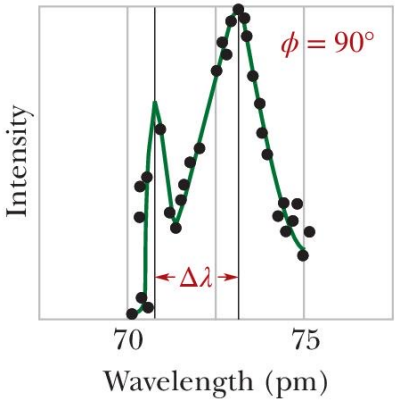
- In 1923 Arthur **Compton** designed an experiment to show that the photons transfer both **Energy** and **Momentum**.

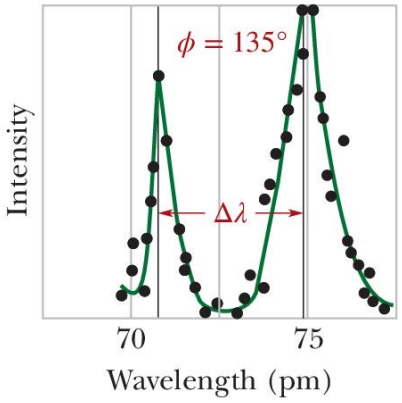


A beam if x-rays of a wavelength $\lambda = 71.1\times10^{-12}m$ is directed onto a carbon target.









Compton's Experiment

- Analogously to the **Photoelectric effect**:

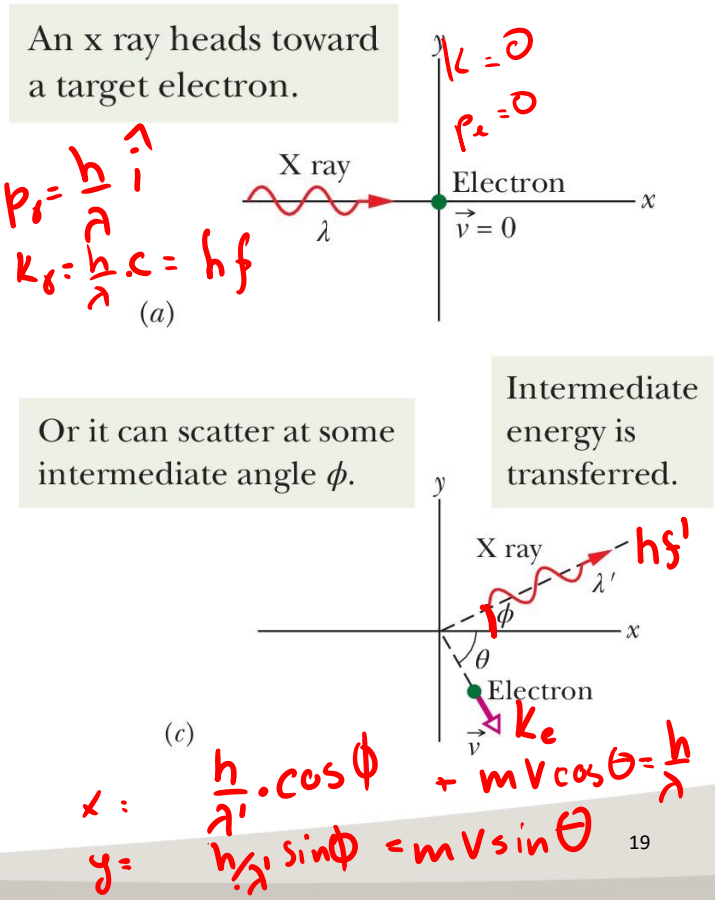
$$hf = hf' + K_e$$

- **Special relativity:** Kinetic energy of the particles is given by

$$K = mc^2(\gamma - 1)$$

- **Energy Conservation:**

$$\frac{h}{\lambda} = \frac{h}{\lambda'} + mc^2(\gamma - 1)$$



Compton's Experiment

- **Momentum of a relativistic Particle:**

$$\vec{p} = \gamma m \vec{v}$$

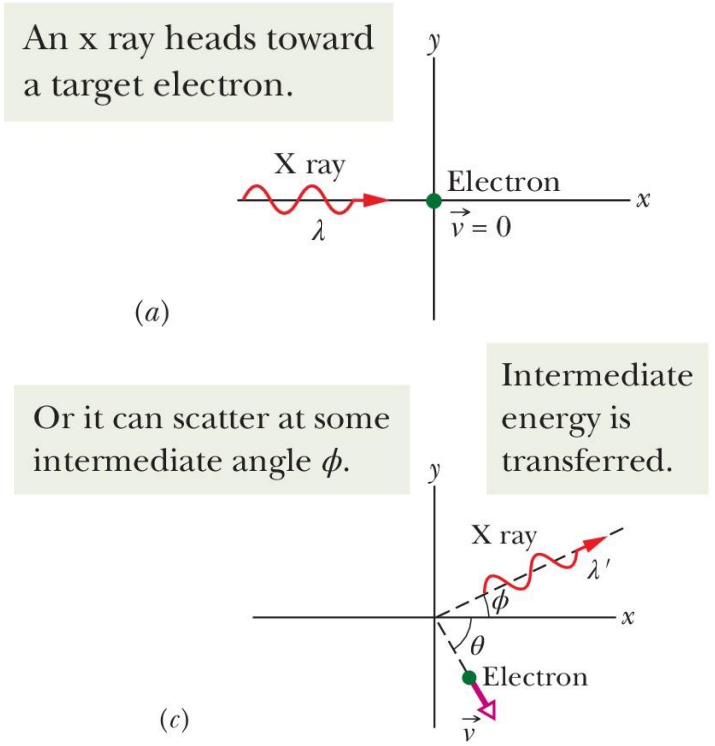
- **Momentum Conservation:**

$$\frac{h}{\lambda} = \frac{h}{\lambda'}\cos\phi + \gamma m v \cos\theta \quad (x\ axis)$$

$$0 = \frac{h}{\lambda'}\sin\phi - \gamma m v \sin\theta \quad (y\ axis)$$

- **Compton Shift:**

$$\Delta\lambda = \frac{h}{mc}(1 - \cos\phi)$$



Question 2

- X-rays with wavelength $\lambda = 22\ \mathrm{pm}$ are scattered from carbon target and the Compton shift of 2.2 pm is observed. What is the fractional energy loss of x-ray photons in the scattering?

A. 0.909

B. 0.01

C. 0.091

D. $24.2pm$

Question 3

- A Compton collision occurs between a photon and a free electron. What angle ϕ will give the maximum wavelength shift for the photon?

A. $\phi = 0$

B. $\phi = \pi/2$

C. $\phi = \pi$

D. $\phi = 3\pi/2$

Next Time

- Particles are also Probability waves if you look hard enough.
- Electrons Experience Diffraction.

